RAJALAKSHMI ENGINEERING COLLEGE

(An Autonomous Institution)

Rajalakshmi Nagar, Thandalam - 602 105

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

(CYBER SECURITY)

CR23532 - CYBER AND DIGITAL FORENSICS

LABORATORY RECORD

SUBMITTED BY

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EXPERIMENT 01

Study of Computer Forensics and Different Tools Used for Forensic Investigation

DATE: 22/07/2026

AIM

To understand the stages of computer forensic investigation and compare the tools used for acquisition, preservation, recovery and examination of digital evidence.

TOOLS / SOFTWARE REQUIRED

EnCase, FTK/FTK Imager, ProDiscover, Autopsy, The Sleuth Kit, USBDeview and other forensic utilities.

OVERVIEW

Computer forensics involves identifying potential evidence, preserving it, acquiring a reliable copy, examining artifacts and presenting findings. Files, device records, browser histories and logs may help reconstruct a digital incident.

PROCEDURE

1. Review the purpose of digital forensics and identify typical evidence sources such as disks, USB devices, memory, user files, system logs and browser profiles.
2. Arrange the investigation workflow in order: identification, collection, preservation, examination, analysis and reporting. Note the importance of documenting each stage.
3. Study commercial investigation suites such as EnCase, FTK and ProDiscover, noting their disk examination, searching and reporting capabilities.
4. Examine open-source applications including Autopsy and The Sleuth Kit, and identify how they support filesystem examination and artifact analysis.
5. Compare FTK Imager with full forensic suites, focusing on imaging, previewing evidence and hash verification rather than case-level analysis.
6. Identify specialist utilities for USB history, activity timelines, browser traces and image metadata. Summarize the function of each tool in a comparison table.
7. Note the basic evidence-handling precautions: authorized access, read-only acquisition where possible, hash recording and a maintained chain of custody.

OBSERVATION

Item Examined	Observation / Significance
EnCase / FTK	Commercial suites for broad evidence examination and reporting.
Autopsy / Sleuth Kit	Disk, filesystem, file and user-activity investigation.
FTK Imager	Creating and verifying forensic disk images.
USBDeview / LastActivityView	Reviewing USB-device and system-activity history.

Item Examined	Observation / Significance
ExifReader	Displaying photographic metadata embedded in supported images.

OUTPUT

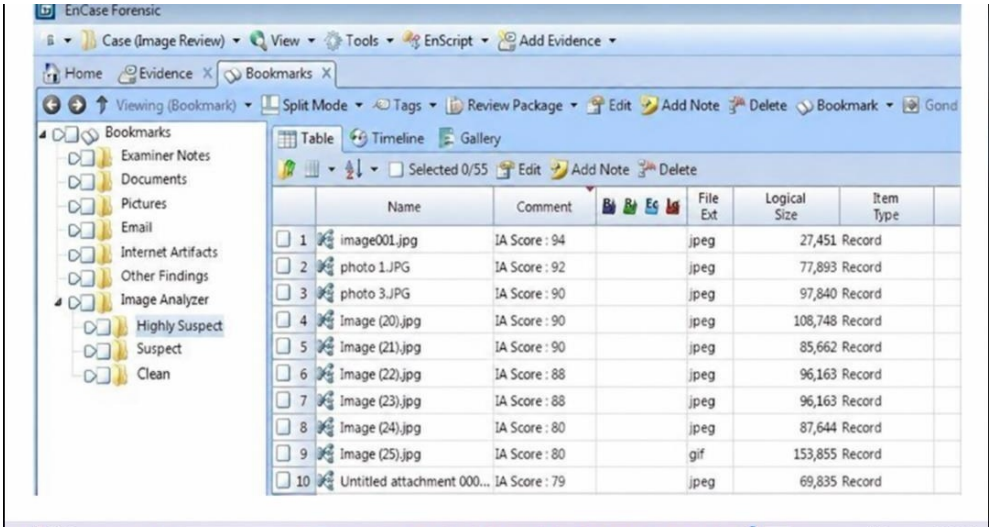


Fig. 1.1 EnCase Forensic interface displaying file evidence

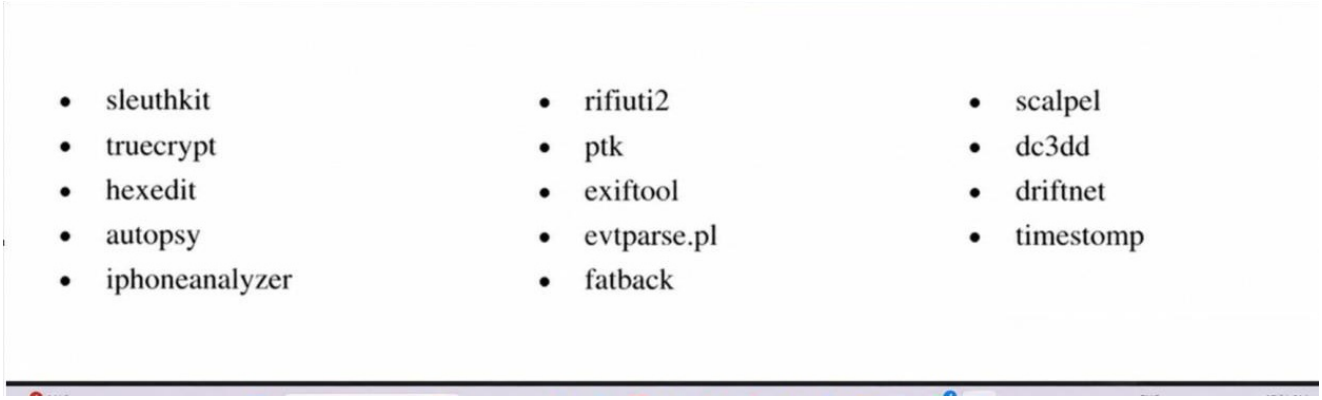


Fig. 1.2 Forensic utilities available in a Linux forensic environment

RESULT

The investigation stages and the purposes of common computer forensic tools were studied and compared.

EXPERIMENT 02

Introduction to Digital Forensics (TryHackMe)

DATE: 29/07/2026

AIM

To learn the fundamentals of digital evidence handling and the stages of an investigation using the TryHackMe Introduction to Digital Forensics room.

TOOLS / SOFTWARE REQUIRED

Web browser, TryHackMe learning platform - Intro to Digital Forensics room.

OVERVIEW

Digital evidence may be present in devices, user activity, communication records and associated storage. A forensic process seeks to recover relevant facts without unnecessarily changing the original material.

PROCEDURE

1. Sign in to TryHackMe and navigate to the Intro to Digital Forensics learning room.
2. Read the opening task and list the kinds of physical devices and digital traces that may be useful at an incident scene.
3. Study the digital forensics process task; distinguish evidence identification and acquisition from subsequent examination and interpretation.
4. Review preservation practices, including controlled access, documentation and avoiding unnecessary changes to the original device.
5. Work through the practical scenario and identify the sources of evidence relevant to the facts of the case.
6. Answer the room questions after reading each task, then verify your understanding using the feedback and explanations.
7. Review the task overview and prepare a concise summary of the six main forensic stages and their intended outcomes.

OBSERVATION

Item Examined	Observation / Significance
Evidence sources	Computers, removable storage, phones and corresponding digital traces.
Forensic workflow	Identification, preservation, collection, examination, analysis and presentation.
Practical scenario	Evidence sources evaluated against an incident narrative.

OUTPUT

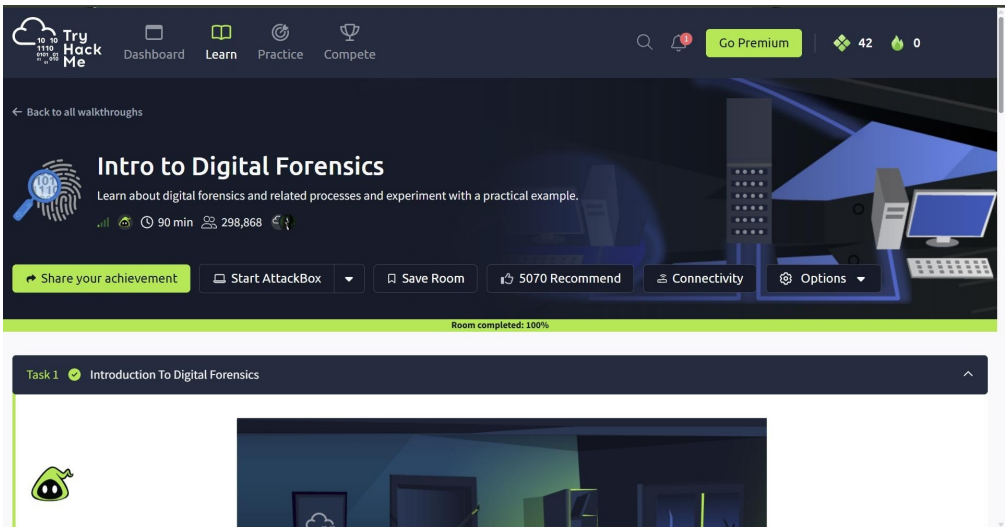


Fig. 2.1 TryHackMe Intro to Digital Forensics learning room

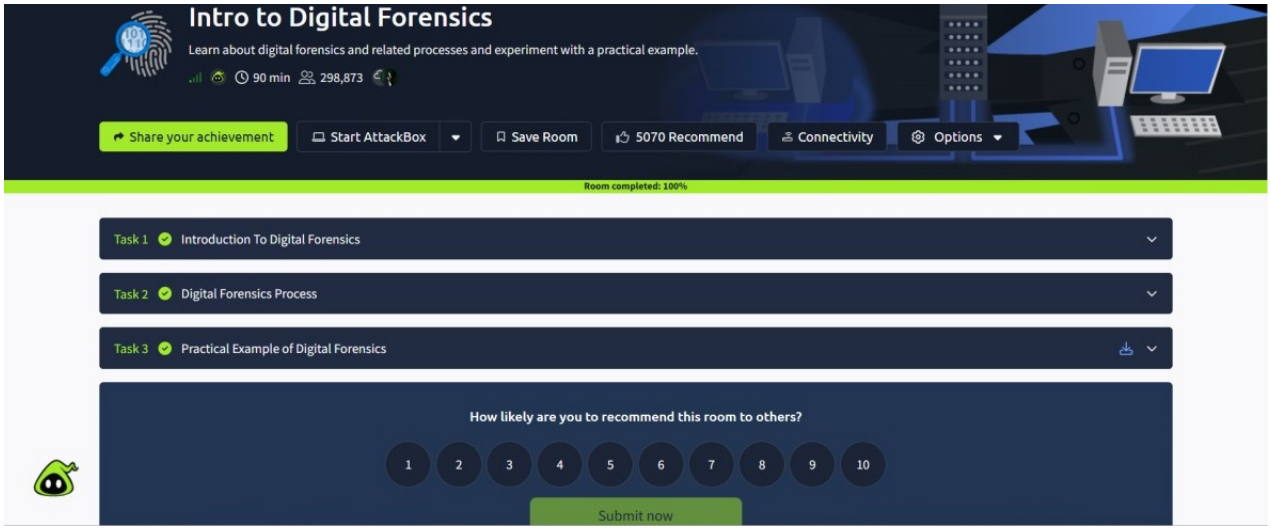


Fig. 2.2 Overview of the completed learning tasks

RESULT

Fundamental digital forensic concepts and the structure of a formal evidence investigation were reviewed through the TryHackMe learning room.

EXPERIMENT 03

Windows Forensics 1 (TryHackMe)

DATE: 05/08/2026

AIM

To study Windows registry artifacts and other operating-system traces used to investigate user activity and attached devices.

TOOLS / SOFTWARE REQUIRED

TryHackMe Windows Forensics 1 room; Windows Registry Explorer/Registry Editor (for study).

OVERVIEW

Windows stores forensic traces in registry hives, recent-file artifacts, Prefetch records, Jump Lists and USB-device entries. Their interpretation requires attention to artifact origin and timestamp meaning.

PROCEDURE

1. Open the Windows Forensics 1 room in TryHackMe and read the introduction to Windows forensic evidence.
2. Identify the main Windows registry hives: SYSTEM, SOFTWARE, SAM, SECURITY, NTUSER.DAT and USRCLASS.DAT.
3. Study how registry hives can be collected from a forensic image and examined offline rather than editing the evidence system.
4. Review system and account artifacts, including computer information, shutdown records and user-account details.
5. Explore traces of file and folder usage such as RecentDocs, ShellBags and Jump Lists, and distinguish them from direct evidence of a file being opened.
6. Review evidence-of-execution sources, including Prefetch files and relevant registry keys, noting that no single artifact establishes every execution event.
7. Examine USB-device forensic artifacts and how identifiers and connection records may assist in linking removable devices to a computer.
8. Complete the learning tasks and record the purpose of each major artifact category.

OBSERVATION

Item Examined	Observation / Significance
Registry hives	Contain configuration and user-specific forensic information.
Execution traces	May include Prefetch records, Run keys and Jump Lists.
External devices	Device identities and connection-related records can be examined.

OUTPUT

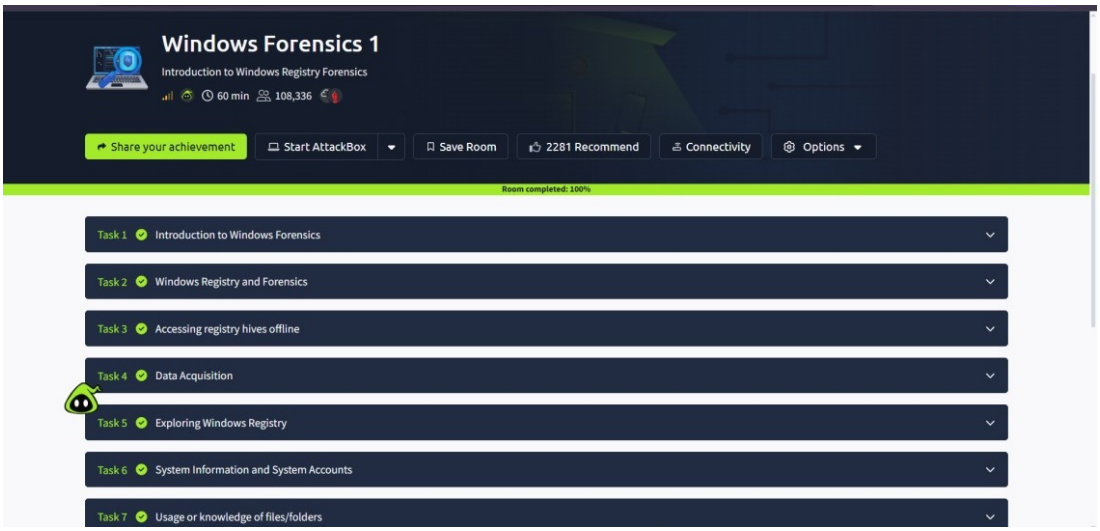


Fig. 3.1 Windows Forensics 1 task list in TryHackMe

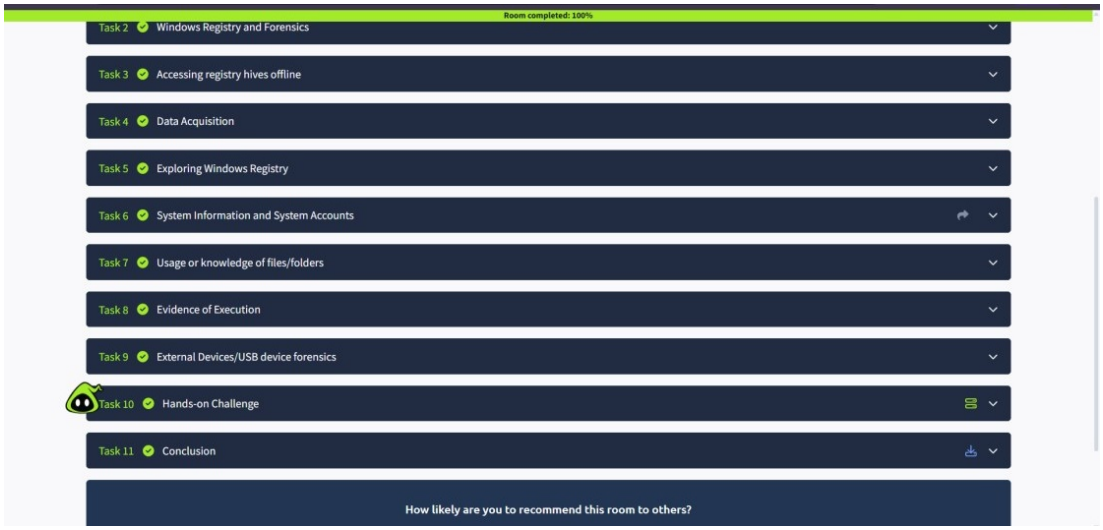


Fig. 3.2 Windows Forensics lesson sections and task status

RESULT

Windows registry hives, file usage artifacts, execution indicators and USB-related traces were studied.

EXPERIMENT 04

Live Forensics Case Investigation Using Autopsy

DATE: 12/08/2026

AIM

To create a forensic case in Autopsy, ingest an authorized data source and inspect user files and system artifacts.

TOOLS / SOFTWARE REQUIRED

Autopsy Digital Forensics Platform; authorized disk image, logical files or test data source.

OVERVIEW

Autopsy groups digital evidence into cases and uses ingest modules to identify files and artifacts. Investigators can tag relevant items and generate reports without relying solely on manual directory browsing.

PROCEDURE

1. Launch Autopsy and choose New Case. Set the case name, case storage directory and case number.
2. Enter the examiner information as requested by the application and finish creating the case database.
3. Choose Add Data Source, select an authorized local drive, disk image or logical-file folder, and confirm the data-source path.
4. Enable suitable ingest modules, such as File Type Identification, Recent Activity, Hash Lookup and Keyword Search, where supported.
5. Run ingest and monitor the task progress until Autopsy finishes processing the available data.
6. Inspect Data Sources, File Views and Results. Review accessible documents, downloads, browser-related traces and any deleted-file entries.
7. Select relevant files to examine filenames, paths, timestamps, type and size; use keyword searching to narrow the results if needed.
8. Tag noteworthy evidence and generate an HTML report from the report wizard; review the report for the expected details.

OBSERVATION

Item Examined	Observation / Significance
Case configuration	Case identity and selected data source are recorded.
Ingest output	Files and supported artifact categories appear in the results tree.
Evidence report	Selected findings can be exported in a readable report.

OUTPUT

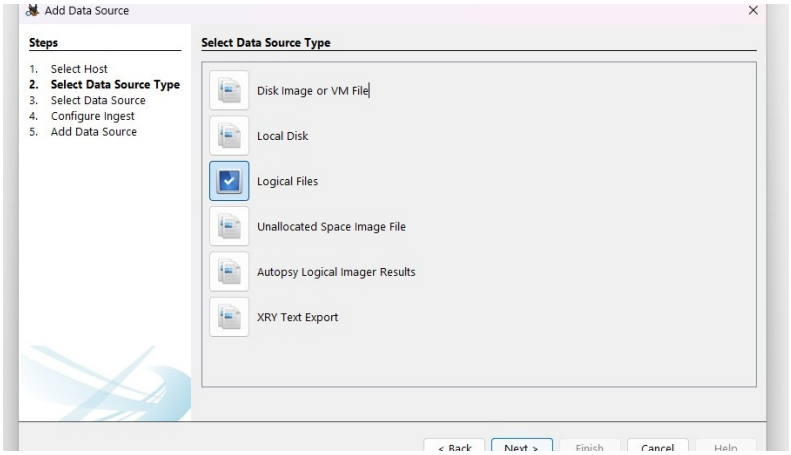


Fig. 4.1 Autopsy data-source selection interface

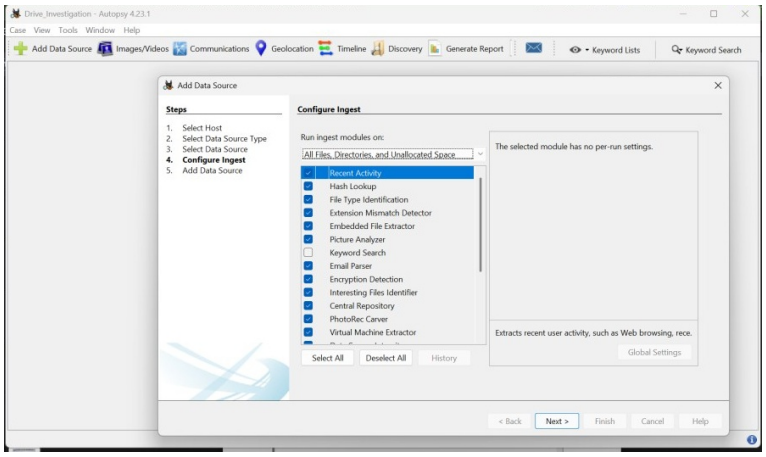


Fig. 4.2 Configuring Autopsy ingest modules

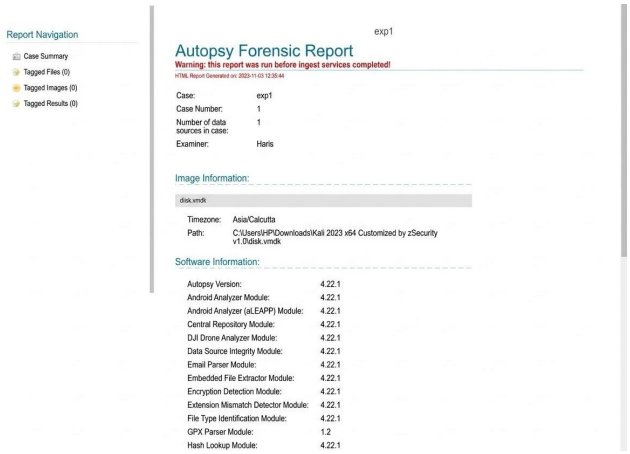


Fig. 4.3 Sample HTML forensic report generated by Autopsy

RESULT

The Autopsy case workflow, ingest process, evidence navigation and reporting functions were examined.

EXPERIMENT 05

Recover Deleted Files Using Forensic Tools (FTK Imager)

DATE: 19/08/2026

AIM

To acquire a test storage source, examine deleted-file evidence and understand recovery and integrity verification.

TOOLS / SOFTWARE REQUIRED

FTK Imager; test USB drive or authorized forensic image; destination folder on a separate storage location.

OVERVIEW

Deleted files may remain partially recoverable until their sectors are overwritten. A forensic image reduces direct handling of the original media, and cryptographic hashes can help document integrity.

PROCEDURE

1. Prepare a disposable test storage device and place sample files on it. Delete one non-sensitive sample file to create a recovery scenario.
2. Open FTK Imager and use File > Create Disk Image; choose the appropriate physical or logical source type.
3. Select and verify the source, then specify an evidence image format such as E01 or RAW/dd.
4. Enter an evidence identifier and choose a destination directory on a different drive, with verification enabled if available.
5. Start acquisition and check that the imaging process completes. Record the generated image name and verification hashes.
6. Add the acquired image as evidence in FTK Imager or another supported forensic tool and inspect the directory structure.
7. Locate the deleted-file entry or recoverable data, then export the available file to an isolated destination folder.
8. Inspect the recovered file and record its name, type, location, recovery status and any available hash values.

OBSERVATION

Item Examined	Observation / Significance
Disk image	An acquired evidence copy is generated in the chosen image format.
Deleted-file entry	A deleted sample may be visible if filesystem metadata or content remains.
Integrity	Acquisition and verification hash values can be compared.

OUTPUT

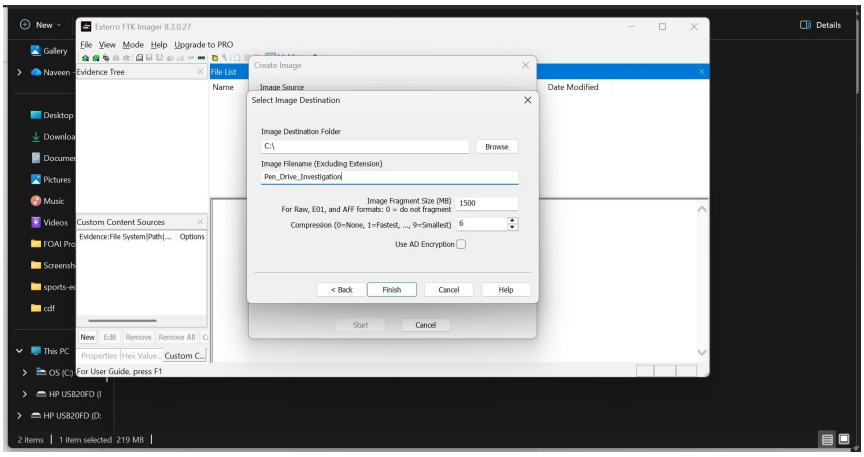


Fig. 5.1 FTK Imager evidence details during acquisition

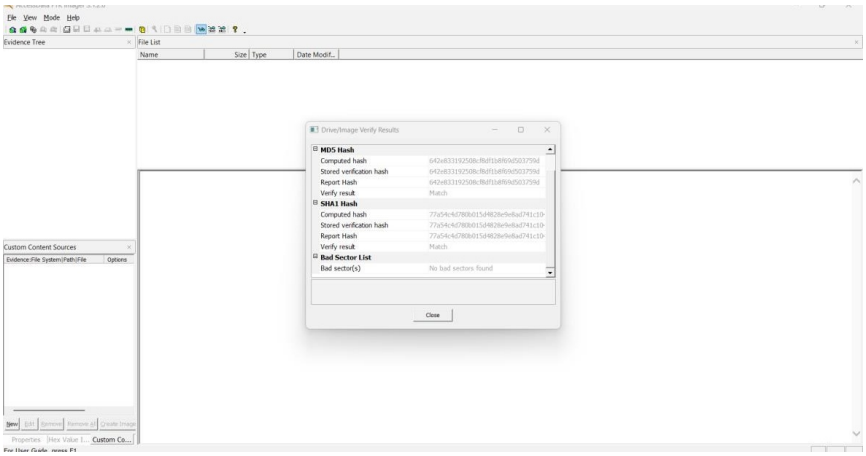


Fig. 5.2 FTK Imager evidence verification output

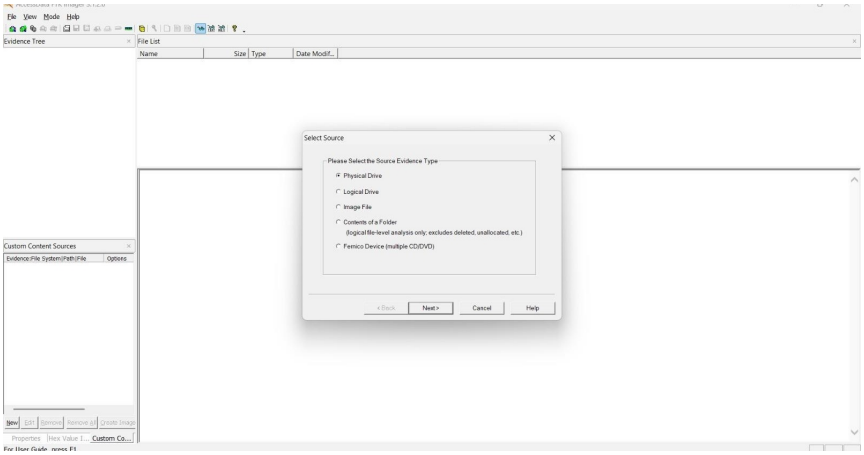


Fig. 5.3 Forensic tool evidence-source selection

RESULT

The procedure for forensic disk imaging, locating recoverable deleted files and checking acquisition integrity was studied.

EXPERIMENT 06

Find the Last Connected USB Device (USB Forensics)

DATE: 02/09/2026

AIM

To inspect current and historical USB connections and identify the most recent relevant removable-storage device.

TOOLS / SOFTWARE REQUIRED

USBDeview (NirSoft); Windows test computer.

OVERVIEW

USB history can provide device description, identifiers and registry-associated times. Interpreting the most recent connection requires checking the meaning of the available fields instead of simply selecting the last displayed row.

PROCEDURE

1. Run USBDeview on an authorized Windows computer and wait for the list of present and previously connected devices.
2. Review the visible fields: device name, description, type, vendor/product identifiers, serial number and connection status.
3. Filter or sort the list to focus on mass-storage devices and examine the relevant connection or registry timestamp columns.
4. If a test USB drive is available, connect it and refresh the displayed list so its new entry can be identified.
5. Open Properties for the selected device and examine its identifiers, drive letter and available time information.
6. Compare the recorded timestamps of relevant devices, noting that registry timestamps may reflect more than one type of event.
7. Document the best-supported most recent USB device and record the evidence fields without disabling or uninstalling it.

OBSERVATION

Item Examined	Observation / Significance
Device identity	Description, type, vendor/product ID and serial number where recorded.
Activity fields	Connected status and available registry/connection times.
Finding	Most recent matching removable-storage record identified from available metadata.

OUTPUT

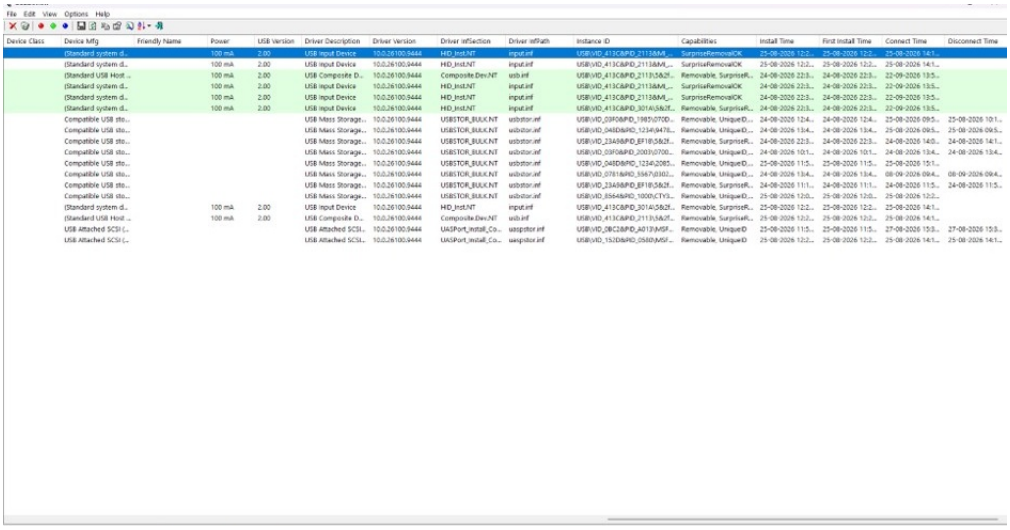


Fig. 6.1 USBDeview list of previously attached USB devices

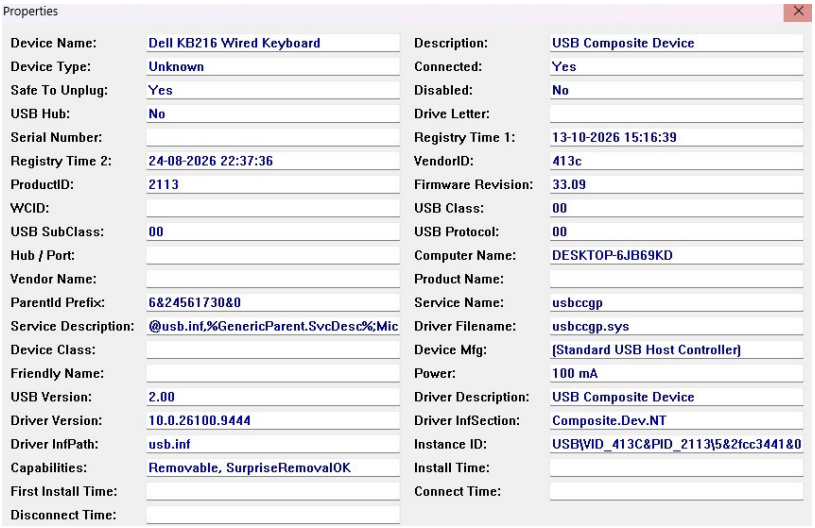


Fig. 6.2 Properties window showing detailed USB device identifiers

RESULT

The USB connection-history inspection procedure was performed on sample records, and the device-identification fields were documented.

EXPERIMENT 07

View the Last Activity of Your PC

DATE: 09/09/2026

AIM

To examine recent application, file and system events with LastActivityView and organize them into a forensic timeline.

TOOLS / SOFTWARE REQUIRED

LastActivityView (NirSoft); authorized Windows test computer.

OVERVIEW

Activity viewers combine records from several Windows sources. A timeline may show file access, program execution and logon-related activity, but the meaning of each event depends on its underlying artifact.

PROCEDURE

1. Download or open LastActivityView and launch the program on the Windows test system.
2. Allow the utility to assemble recent activity records from supported sources, including relevant registry and system traces.
3. Inspect columns such as action time, description, file or folder path, event type and data source.
4. Sort the results by action time so the newest supported activities can be reviewed first.
5. Open the properties of one or more selected records and note the event details and the source that supplied each entry.
6. Identify examples of application launches, opened files or folders, browser-related events and available system activities.
7. Record a small event timeline and distinguish observed artifacts from conclusions that would require corroboration.

OBSERVATION

Item Examined	Observation / Significance
Action time	Timestamp associated with the identified artifact.
Activity description	Program, file, folder or operating-system event.
Data source	Underlying record type used to assemble the timeline.

OUTPUT

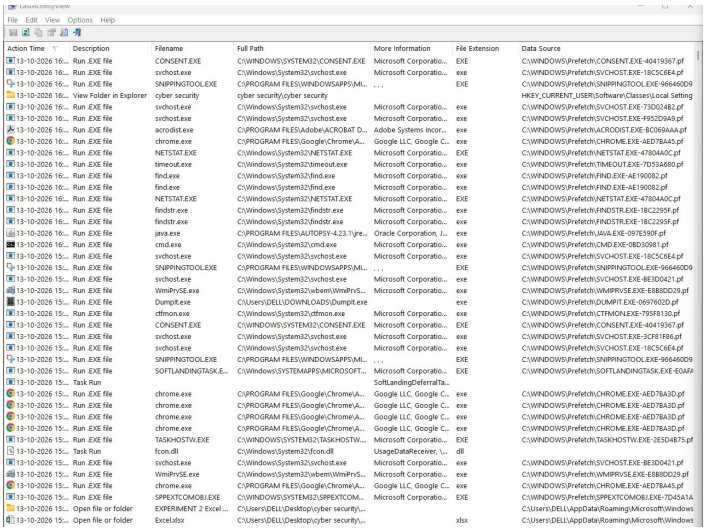


Fig. 7.1 LastActivityView timeline of user and system events

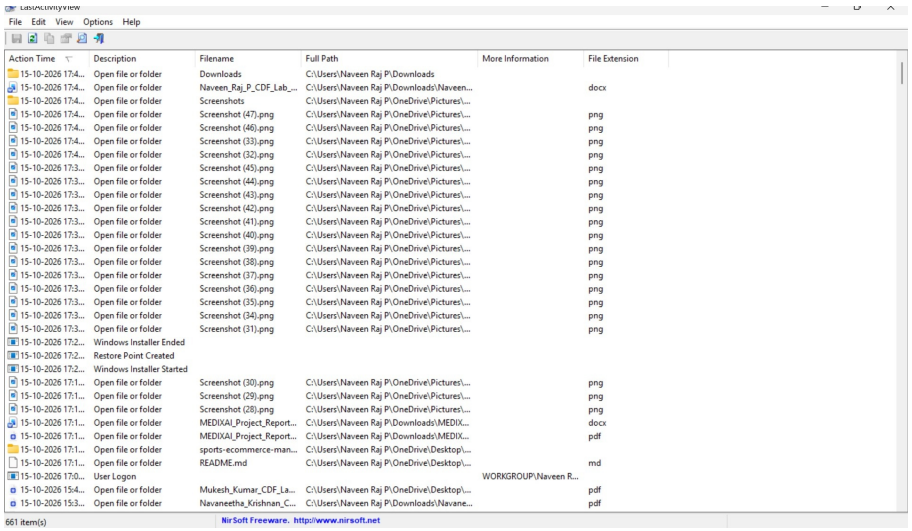


Fig. 7.2 Recent activity records shown in LastActivityView

RESULT

Recent PC activities were reviewed and organized using the timestamps and descriptions shown by LastActivityView.

EXPERIMENT 08

Extraction of Browser Artifacts

DATE: 16/09/2026

AIM

To examine browsing-history and cache artifacts using ChromeHistoryView and related browser forensic utilities.

TOOLS / SOFTWARE REQUIRED

ChromeHistoryView, ChromeCacheView; a browser profile from an authorized Windows system.

OVERVIEW

Browser artifacts may include visited URLs, page titles, visit times, download records and cached resources. Results depend on the browser version, user profile, retention settings and any deletion of browser data.

PROCEDURE

1. Run ChromeHistoryView for the authorized Chrome profile or select a saved copy of the browser history database.
2. Inspect the history entries and note the URL, title, visit date and time, visit count and available referrer data.
3. Sort the records by the last-visit field and use the search/find function for a keyword or website of interest.
4. Open Properties for a selected entry to review its source profile and detailed visit information.
5. Run ChromeCacheView or another suitable cache examiner against the matching profile where supported.
6. Review any accessible cached URL, MIME/content type, file size and timestamp details.
7. Compare the two evidence categories and document the observations, including browser-version or retention limitations.

OBSERVATION

Item Examined	Observation / Significance
History	Visited site address, title and available visit timestamps.
Search/filter	Matching entries found for a selected keyword.
Cache	Cached resources and metadata when present in the profile.

OUTPUT

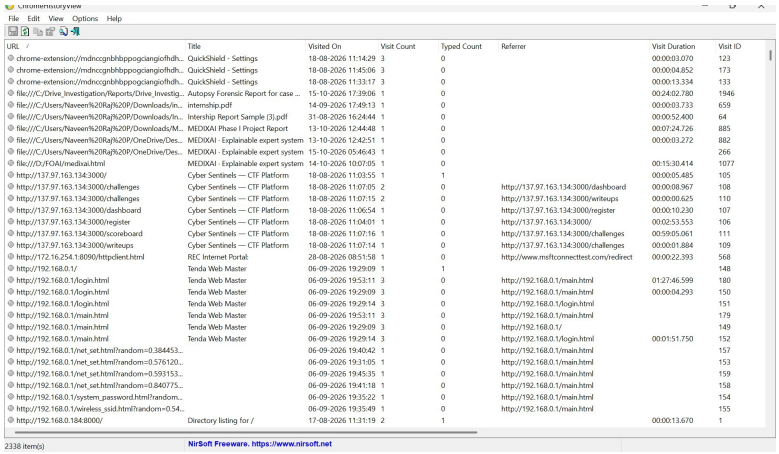


Fig. 8.1 ChromeHistoryView results for browser history

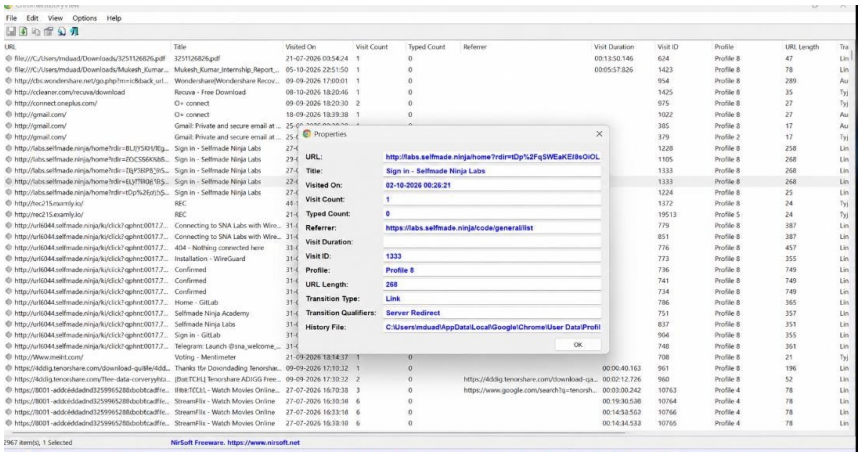


Fig. 8.2 Properties of an individual browser-history record

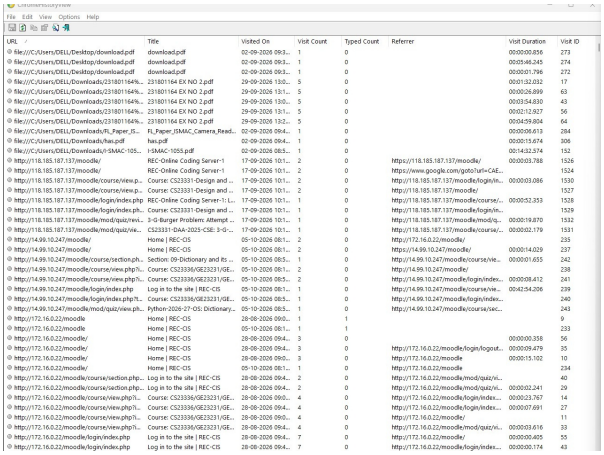


Fig. 8.3 Browser artifact list in the history viewer

RESULT

Browser history and related artifacts were examined using the available forensic browser-viewer utilities.

EXPERIMENT 09

Comparison of Two Files Using Compare It Software

DATE: 23/09/2026

AIM

To compare two versions of a text file and identify additions, deletions and modified lines.

TOOLS / SOFTWARE REQUIRED

Compare It; two prepared sample text files.

OVERVIEW

File comparison helps investigators document changes between an original file and a modified version. A visual diff shows textual changes but should not be treated as a cryptographic integrity check.

PROCEDURE

1. Create a baseline text file with several lines of sample content and save a second copy under a different name.
2. Make controlled changes to the second file, such as adding one line, deleting another and altering selected words.
3. Launch Compare It and open the baseline file in the left comparison pane.
4. Open the modified file in the right pane and start the comparison.
5. Study highlighted insertions, deletions and modified sections and use navigation buttons to move between differences.
6. Verify the changes against the original sample files and note the affected lines and change types.
7. Generate or save the comparison report if the installed software supports this feature.

OBSERVATION

Item Examined	Observation / Significance
Source file	Original version used as the comparison baseline.
Changed file	Controlled added, removed and edited text.
Difference report	Visual indicators and a list of detected changes.

OUTPUT

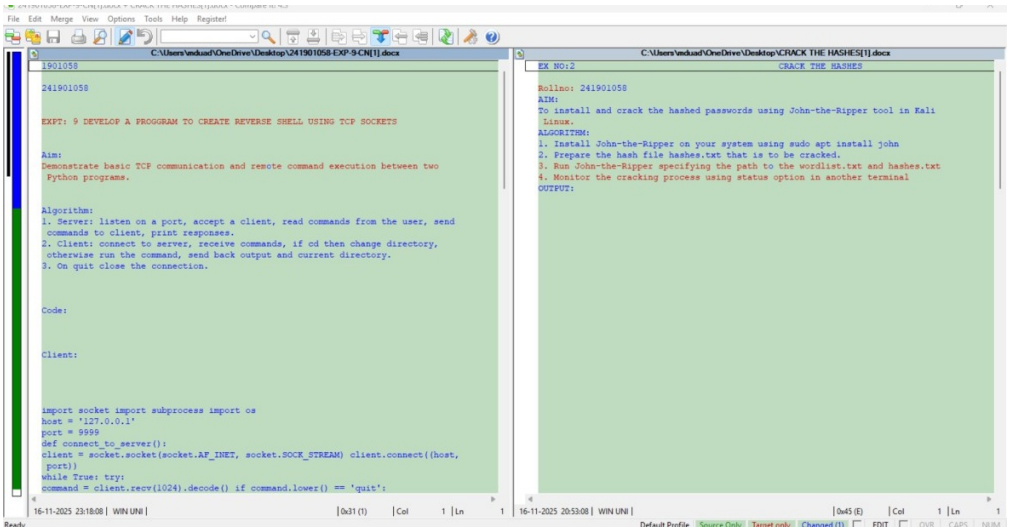


Fig. 9.1 Compare It visual comparison of two edited files

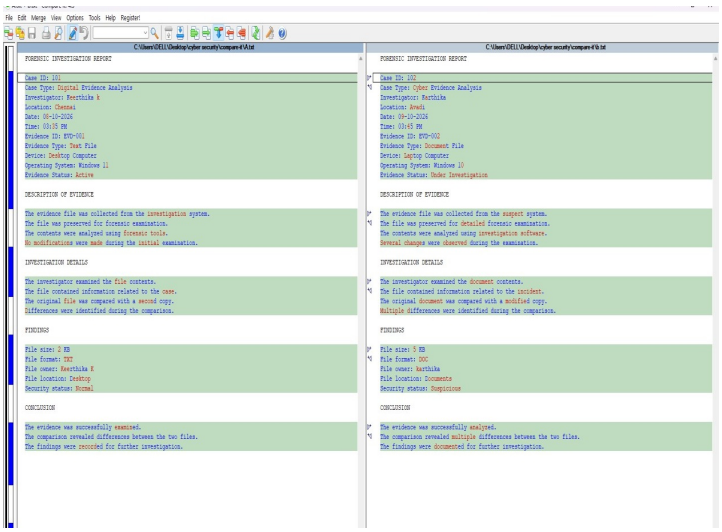


Fig. 9.2 Differences highlighted side by side in Compare It

RESULT

The two sample files were compared, and differences between their text contents were identified.

EXPERIMENT 10

Collecting Email Evidence from a Victim PC

DATE: 30/09/2026

AIM

To understand the acquisition of volatile memory and the extraction of potential email-address artifacts for a forensic examination.

TOOLS / SOFTWARE REQUIRED

DumpIt and Bulk Extractor / Bulk Extractor Viewer; authorized test-system memory capture.

OVERVIEW

Memory images can contain strings and other artifacts associated with recent activities. Email-address strings extracted from memory are leads only; their presence alone does not prove that an email was sent or received.

PROCEDURE

1. Prepare an authorized test machine and a separate destination with sufficient free space to receive a RAM image.
2. Run DumpIt with the required permissions and confirm the memory acquisition prompt.
3. Allow the tool to save the physical memory image and record the resulting file location and size.
4. Open Bulk Extractor or its viewer and select the captured memory image as the input evidence.
5. Choose an output directory and start feature extraction using the available scanners.
6. Inspect generated feature files such as email.txt, email_histogram.txt and url.txt, if produced.
7. Review potential addresses together with offsets, source context and histogram counts; exclude obvious irrelevant strings.
8. Document the evidence image and extracted artifacts. Treat all candidate email evidence as subject to further validation.

OBSERVATION

Item Examined	Observation / Significance
Memory capture	Raw or comparable acquired image used as extraction input.
Email features	Email-like strings recovered from available data.
Related artifacts	Feature histograms and URLs, when the respective scanners return results.

OUTPUT

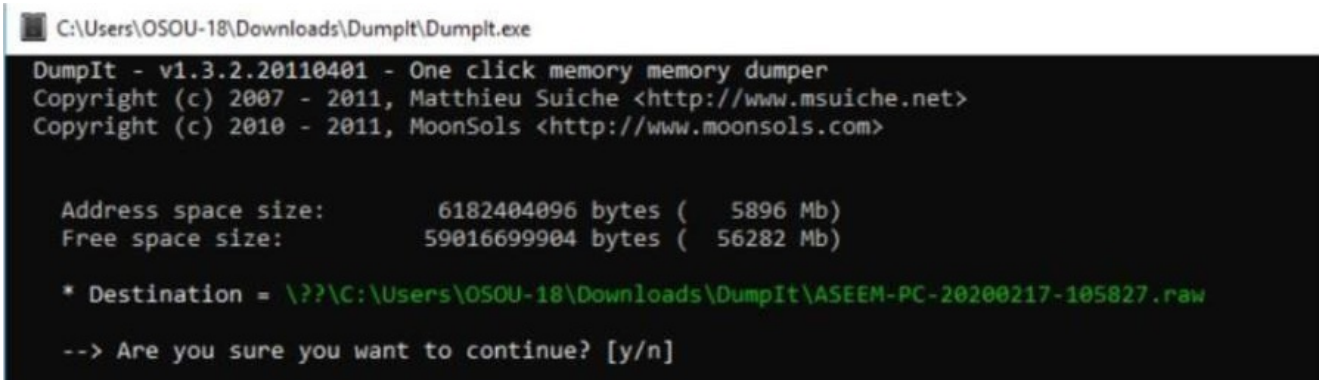


Fig. 10.1 DumpIt memory acquisition command window

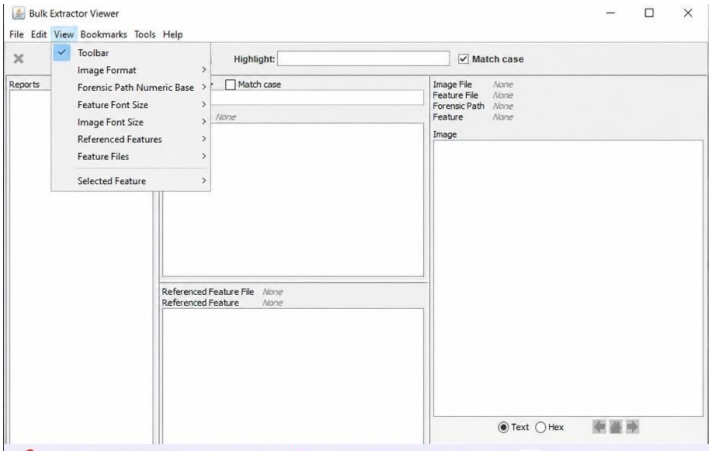


Fig. 10.2 Bulk Extractor Viewer listing extracted feature categories

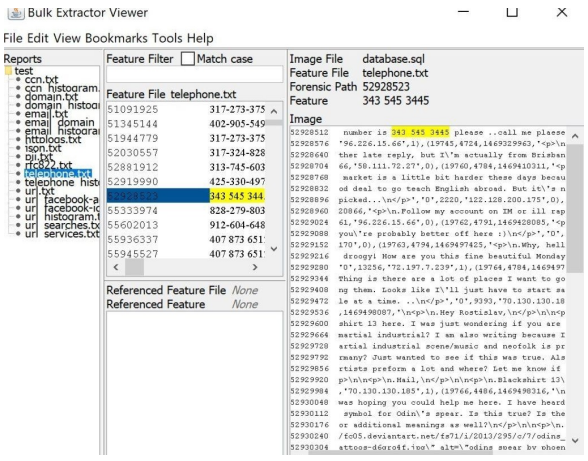


Fig. 10.3 Extracted features and evidence offsets in Bulk Extractor

RESULT

The memory-capture and email-feature extraction workflow was studied, and sample extracted artifact categories were examined.

EXPERIMENT 11

Hiding and Extracting Text Behind an Image or Audio File Using CMD

DATE: 07/10/2026

AIM

To demonstrate simple file concatenation that appends text or archive data to a carrier file and to inspect the resulting output.

TOOLS / SOFTWARE REQUIRED

Windows Command Prompt, Notepad, test JPG/MP3 file and sample TXT/ZIP files.

OVERVIEW

Binary concatenation can append data to the end of an image or audio file. The carrier may still display or play, but this simple technique is not encryption and does not guarantee concealment from forensic examination.

PROCEDURE

1. Create a sample text file and copy it, together with a test JPEG, into an isolated working directory.
2. Open Command Prompt and change the current folder to the directory containing the test files.
3. Use `copy /b image.jpg + secret.txt combined.jpg` to append the text bytes to the JPEG without changing the source files.
4. Open combined.jpg in an image viewer to verify whether the format parser still displays the image.
5. Inspect the combined file with a text or hexadecimal viewer and search near its end for the appended text.
6. Optionally repeat the test with a ZIP archive and a copy of an MP3 file using the same binary concatenation concept.
7. Document the output filenames, commands, change in file sizes and the method used to retrieve the appended content.
8. Conclude by noting that appended data can often be discovered by inspecting file structure and trailing bytes.

OBSERVATION

Item Examined	Observation / Significance
Input files	A carrier image/audio file and a separate test payload.
Combined file	A new file with the appended bytes and larger overall size.
Inspection	Payload bytes visible through suitable file inspection tools.

OUTPUT

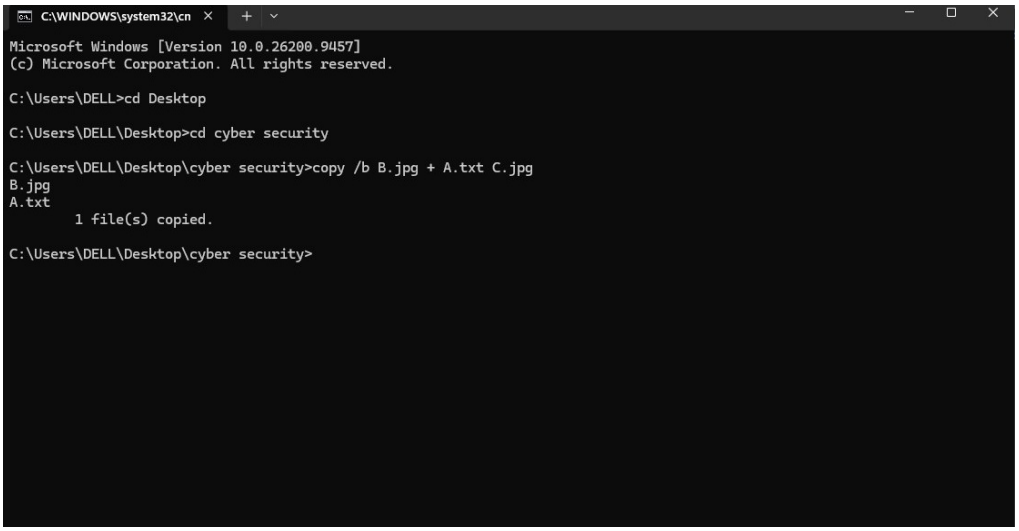


Fig. 11.1 Command Prompt showing binary copy and merged-file output

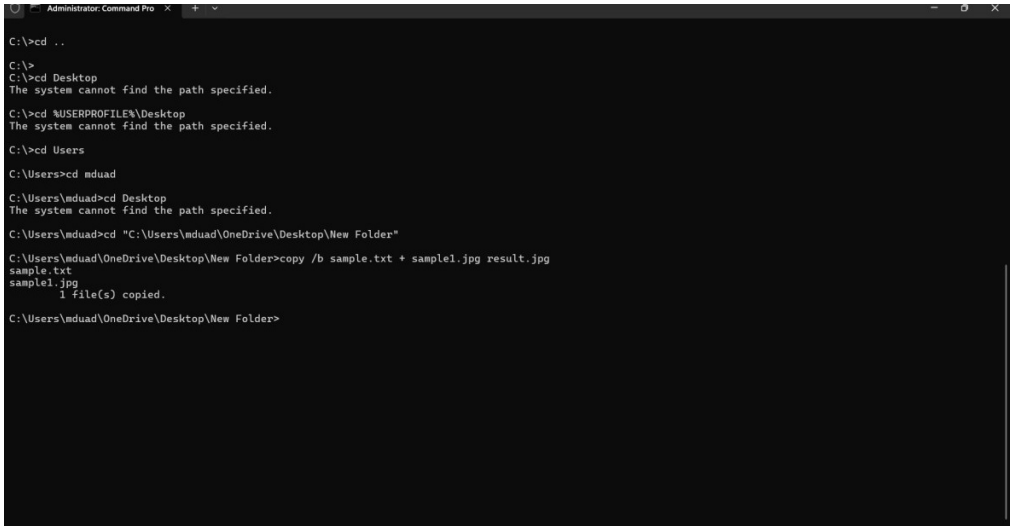


Fig. 11.2 Command-line steps for combining a carrier and a text file

RESULT

Binary concatenation of a test carrier and payload was illustrated, and the approach for inspecting the appended data was understood.

EXPERIMENT 12

Extraction of EXIF Data from Image Files Using ExifReader

DATE: 14/10/2026

AIM

To examine image metadata such as capture time, camera identification and photographic settings using ExifReader.

TOOLS / SOFTWARE REQUIRED

ExifReader software; a sample JPEG image containing EXIF metadata.

OVERVIEW

EXIF metadata may store camera make/model, capture date, exposure settings and sometimes GPS coordinates. Fields may be absent or edited, so metadata should be corroborated before drawing investigative conclusions.

PROCEDURE

1. Obtain a JPEG test image with metadata and preserve an unchanged copy for reference.
2. Launch ExifReader and use its Open command to select the test image.
3. Confirm that the image preview is displayed and check whether EXIF fields are available.
4. Record the camera manufacturer and model, original image date/time, file dimensions and software tags where present.
5. Review photographic settings such as exposure, aperture, focal length, ISO and flash mode if supplied in the metadata.
6. Check for GPS tags but do not assume that geolocation information must be present in every photograph.
7. Prepare a metadata summary and note which values are absent or require independent verification.

OBSERVATION

Item Examined	Observation / Significance
Device information	Camera make and model where embedded.
Capture fields	Date/time, image dimensions and photographic settings.
Location fields	Optional GPS coordinates if the device recorded them.

OUTPUT



Fig. 12.1 ExifReader displays a photo and associated metadata

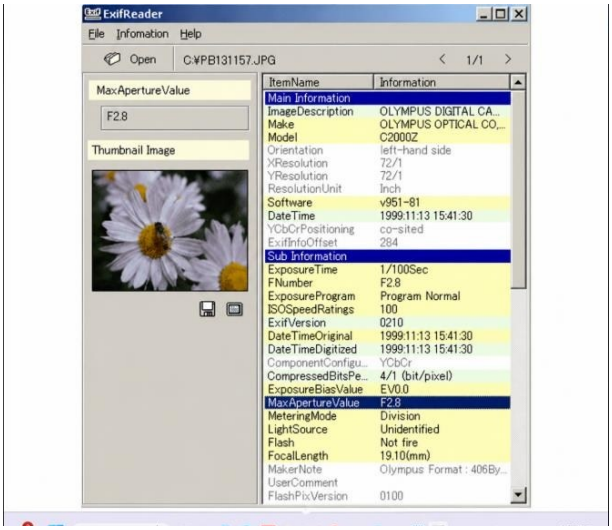


Fig. 12.2 Detailed EXIF metadata fields in ExifReader

RESULT

The method for extracting and interpreting EXIF image metadata using ExifReader was reviewed.